Experiment with functions and relations using digital tools, making and testing conjectures and generalising emerging patterns

Learning intention

- experiment with functions and relations using digital tools, making and testing conjectures and generalising emerging patterns.

Teach and model

- applying transformations to the graph of x^2+y^2=1.
- identifying intervals on the real number line over which a given quadratic function is positive or negative.
- using a table of values to determine when an exponential growth or decay function exceeds or falls below a given value.

Check understanding

- Explain the main idea and give one accurate example.
- Show the idea in a second way: with words, a model, a diagram or symbols.
- Apply the idea to a short unfamiliar situation and justify the result.

Teaching cautions

- Ask the student to explain their choice, not only state an answer.
- Check that key vocabulary is understood in a new example.
- Mix representations so the skill is transferred rather than copied.

Quick lesson sequence

- Connect to prior knowledge and introduce the key vocabulary.
- Model one clear example, then solve a second example together.
- Finish with an independent check and a short student explanation.